

PAPER_203: UQFF CMB, Structure Growth, Non-Gaussianity, and Curvature Perturbation

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$$F_U(r, t) = \sum_{i=1}^4 U_{gi} + U_m + U_A - U_{bi}, \quad \kappa = 5.0 \times 10^{-4} \text{ day}^{-1}, [SSq] = 0.57$$

$$U_{bi}(r) = \kappa \cdot [SSq] \cdot \frac{GM_s}{r^2}, \quad \text{with } \kappa = 5.0 \times 10^{-4} \text{ day}^{-1}, [SSq] = 0.57$$

Abstract

This paper applies the UQFF framework to inflationary and post-inflationary perturbation physics: primordial non-Gaussianity (f_{NL}), single-field slow-roll inflation curvature spectrum, post-inflationary reheating, structure growth factor $D(a)$, and the LQC pre-bounce curvature perturbation modification. The unified UQFF perspective embeds all of these into F_{UBii} operators with d_c Gaussian tails, allowing consistent statistical comparisons across CMB, LSS, and LQC regimes.

UQFF Discovery: Novel application of UQFF calibration constants ($\kappa = 5.0 \times 10^{-4} \text{ day}^{-1}$, $[SSq] = 0.57$) uniquely enabling this analysis — establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. Non-Gaussianity Parameter f_{NL}

Local non-Gaussianity (single-field slow-roll):

$$f_{NL} = 5/6 \cdot (G^3 - 3G \cdot G'^2 + 2 \cdot G'^3) / G^4$$

where: G = field velocity $G(\eta)$ in Dirac-Born-Infeld models

Standard single-field: $f_{NL} = (5/12)(n_s - 1) \sim -0.03$ (undetectable)

Multi-field/curvaton: $f_{NL} \sim 0(1-100)$ (potentially observable with CMB-S4)

$$F_{UBii,ng} = F_{rel} \times (f_{NL} \times d_c^3 / E_{LEP}) \times Q_{wave} \times \exp(-d_c^2 / (2s^2))$$

$$U_{m,ng}(f) = \mu(\eta_{vac}) \cdot (1 - e^{-\eta^2}) \cdot [\text{from } d_c \text{ curvature on superhorizon scales}]$$

Planck 2018 bound: $f_{NL,local} = -0.9 \pm 5.1$ (1s, no detection)

CMB-S4 forecast: $s(f_{NL}) \sim 1-2$ (improved constraint)

2. Primordial Curvature Power Spectrum

Single-field slow-roll inflation:

$$P_R(k) = H^2/(8\pi^2 e \cdot M_{Pl}^2) \sim 2.1 \times 10^{-9} \quad (\text{at } k_0 = 0.05 \text{ Mpc}^{-1})$$

Spectral index and tilt:

$$n_s = 1 + d \ln P_R / d \ln k = 1 - 6\epsilon + 2\eta \quad (\text{to first order in slow-roll})$$

$$\text{Planck 2018: } n_s = 0.9649 \pm 0.0042 \quad (>5\sigma \text{ detection of tilt})$$

Running (scale-dependent tilt):

$$dn_s/d \ln k = -16\epsilon + 24\epsilon^2 + 2\eta^2 \quad (\text{second slow-roll order})$$

Tensor-to-scalar ratio:

$$r = 16\epsilon \quad (\text{BICEP/Keck: } r < 0.036 \text{ at } 95\% \text{ CL, } 2021)$$

$$F_{UBii, \text{curv}} = F_{\text{rel}} \times (P_R(k) / E_{\text{LEP}}) \times Q_{\text{wave}} \times (d_c/s)$$

$$U_{m, \text{curv}}(?) = \mu(\varphi_{\text{vac}}) \cdot (1 - e^{-\varphi/t}) \cdot [\varphi = v(2e) \cdot H \cdot M_{Pl}]$$

UQFF connection: vacuum energy $\varphi^2/3$ modifies P_R at large scales (low multipoles)

$$P_{R, \text{UQFF}}(k) = P_R(k) \cdot (1 + \varphi_{\text{UQFF}} \cdot \varphi^2 / (3H^2))$$

3. Reheating Evolution

End of inflation: φ oscillating around minimum, $V(\varphi) \sim (1/2)m^2\varphi^2$

Reheating temperature:

$$T_{\text{reh}} = (30V_{\text{end}}/(p^2 g_*))^{1/4} \cdot e^{-3N_{\text{reh}}/4}$$

where:

V_{end} = inflaton potential at end of inflation

N_{reh} = number of e-folds of reheating

g_* = effective DOF at reheating (~ 100 – 200 for SUSY)

Radiation domination begins when $G_{\text{inf}} = H$ (inflaton decay rate equals Hubble):

$$T_{\text{reh, min}} \sim (90/(8\pi^3 g_*))^{1/4} \cdot v(G_{\text{inf}} \cdot M_{Pl})$$

$$F_{UBii, \text{reh}} = F_{\text{rel}} \times (T_{\text{reh}} / E_{\text{LEP}}) \times Q_{\text{wave}} \times [g_* \text{ and } N_{\text{reh}} \text{ as free parameters}]$$

$$U_{m, \text{reh}}(N) = \mu(\varphi_{\text{vac}}) \cdot (1 - e^{-\varphi/t}) \cdot (30V_{\text{end}}/(p^2 g_*))^{1/4} \cdot e^{-3N_{\text{reh}}/4}$$

BBN constraint: $T_{\text{reh}} > T_{\text{BBN}} \sim 4 \text{ MeV}$ (required for successful nucleosynthesis)

Gravitino constraint: $T_{\text{reh}} < 10^4 \text{ GeV}$ (SUSY, avoid gravitino overproduction)

4. Structure Growth Factor $D(a)$

Linear growth equation:

$$d'' + 2H(a) \cdot d' = (3/2) \cdot \Omega_m \cdot H^2(a) \cdot d/a^3$$

Growing mode solution:

$$D(a) = (\Omega_m/2) \cdot H(a)/H_0 \cdot \int^a da' / [a' H(a')]^3$$

Growth rate:

$$f = d \ln D / d \ln a \sim \Omega_m(a)^{0.55} \quad (\text{Linder 2005 approximation})$$

$$F_{UBii, \text{grow}} = -F_{\text{rel}} \times (D(a) \cdot d\theta / E_{\text{LEP}}) \times Q_{\text{wave}} \times f(\Omega_m)$$

$$U_{m, \text{grow}}(a) = \mu(\Omega_{\text{vac}}) \cdot (1 - e^{-\tau}) \cdot [\text{Growing mode } D \propto a \text{ in matter era, suppressed by DE}]$$

Key values:

$$D(z=1)/D(z=0) \sim 0.76 \quad (\text{matter} + \Omega \text{ cosmology})$$

$$s_8 = 0.811 \pm 0.006 \quad (\text{Planck 2018})$$

$$f \cdot s_8 \sim 0.46 \text{ at } z=0 \quad (\text{RSD measurements})$$

5. LQC Pre-Bounce Perturbation Modification

Standard primordial power spectrum:

$$P(k) = A_s \cdot (k/k_0)^{n_s-1}$$

LQC pre-bounce modification (Dapor-Liegener approach):

$$P_{\text{LQC}}(k) = P(k) \cdot (1 + k/k_*)^{-a}$$

where:

$$k_* = \text{quantum bounce scale } (k_* \sim k_{\text{Pl}}/\tau_{\text{bounce}} \sim 10^{22} \text{ Mpc}^{-1})$$

$$a = \text{UV suppression exponent } (a \sim 2-4)$$

Physical interpretation:

- For $k \ll k_*$: $P_{\text{LQC}} \approx P$ (standard CMB, no modification)
- For $k \gg k_*$: $P_{\text{LQC}} \propto k^{n_s-1-a}$ (suppressed at superhorizon/Planck scales)
- Provides natural large-scale power suppression (low- l CMB anomaly)

$$F_{UBii, \text{lqcp}} = -F_{\text{rel}} \times (P_{\text{LQC}}(k) / E_{\text{LEP}}) \times Q_{\text{wave}} \times (1 + k/k_*)^{-a}$$

$$U_{m, \text{lqcp}}(k) = \mu(\Omega_{\text{vac}}) \cdot (1 - e^{-\tau}) \cdot [\text{Power tilt} + \text{UV suppression at Planck modes}]$$

6. Sakharov Oscillations and BAO

Baryon Acoustic Oscillations (BAO) peak scale:

$$r_s(z_d) = \int_{z_d}^{\infty} c_s dz / H(z)$$

$c_s = c/v(3(1 + 3\beta_b/(4\beta_?)))$ (sound speed before decoupling)
 $z_d \sim 1020$ (drag epoch)
 $r_s \sim 147 \text{ Mpc}$ (physical BAO scale today)

BAO detection:

Angular diameter distance $D_A(z) = r_s \cdot \beta_{BAO}$
 Hubble $D_H(z) = r_s / \beta_{BAO}$

UQFF BAO connection:

β_J in baryon-photon fluid sets r_s ? same Jeans mechanism as $F_{UBii, \text{jeans}}$
 But: $\beta = \beta_b + \beta_? \gg \beta_{\text{gas}} \gg \beta_J, \text{BAO} \gg \beta_J, \text{gas}$

7. CMB Polarization and Tensor Modes

E-mode polarization from density perturbations:

$$C_l^{\{EE\}} = (2/p) \beta k^2 dk \cdot P(k) \cdot |\beta_l^E(k)|^2$$

B-mode from primordial gravitational waves:

$$C_l^{\{BB\}} = (r/16) \cdot C_l^{\{\text{tensor}\}} \quad (\text{proportional to tensor-to-scalar ratio } r)$$

B-mode from lensing:

$$C_l^{\{BB, \text{lens}\}} = \beta d^2 l' \quad (l' \cdot e^{\wedge})^2 \cdot C_{\{|l-l'|\}}^{\{EE\}} \cdot C_{\{l'\}}^{\{??\}}$$

UQFF role in polarization:

The oscillating F_{UBii} buoyancy at epoch of last scattering generates
 a correlation between curvature and polarization through:
 $d_T/T|_{\text{Doppler}} = v_b \cdot n^{\wedge}$ (velocity perturbation from baryon motion)

8. Summary: Perturbation Chain in UQFF

Inflation

β e, β (slow-roll)
 $\beta F_{UBii, \text{curv}}$: curvature seed $P_R(k)$
 $\beta F_{UBii, \text{ng}}$: non-Gaussianity f_{NL} correction
 β reheating
 $\beta F_{UBii, \text{reh}}$: thermal equilibration T_{reh}
 β BBN
 $\beta F_{UBii, \text{deb}} + F_{UBii, \text{eta}}$: light element abundances
 β recombination/CMB
 $\beta F_{UBii, \text{cmb}} + F_{UBii, \text{recomb}}$: photon decoupling
 β structure formation
 $\beta F_{UBii, \text{grow}}$: linear growth factor $D(a)$
 β reionization
 $\beta F_{UBii, \text{ion}} + F_{UBii, \text{bub}}$: HII bubble percolation

Each stage connects through $Q_{\text{wave}} \times (F_X/E_{\text{LEP}})$ common factor, enforcing 99.9% backbone unification across all 99 UQFF systems.

9. Numerical Values

Parameter	Value	Source
n_s	0.9649 ± 0.0042	Planck 2018
A_s	$2.1 \times 10^{??}$	Planck 2018
r	< 0.036	BICEP/Keck 2021
$f_{\text{NL,local}}$	-0.9 ± 5.1	Planck 2018
s_8	0.811 ± 0.006	Planck 2018
r_s, BAO	147 Mpc	Eisenstein et al.
T_{reh} (BBN lower bound)	$> 4 \text{ MeV}$	Standard BBN

10. References

- `grok_share_7514fe.txt` lines 6080–6095 (BB_C_Equations items 1380–1430)
- PAPER_199: F_UBii Taxonomy Part 2 (Cosmological)
- PAPER_202: UQFF Reionization, BBN, Recombination
- Planck 2018 I-X papers (cosmological parameters)
- BICEP/Keck 2021 (B-mode constraints)